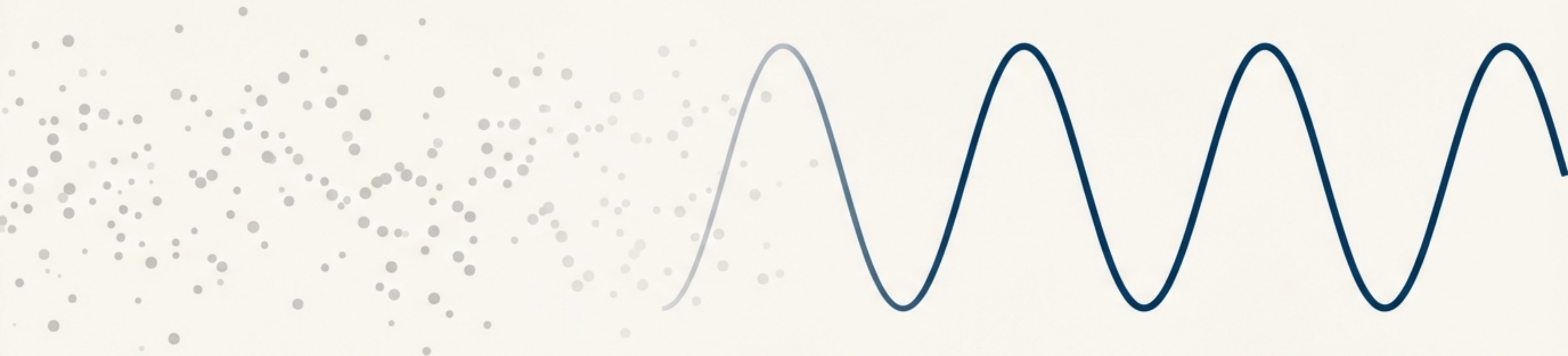




Making Sense of an Uncertain World

How to Separate True Signals from Random Noise

Every day, we face a fundamental challenge: Is what we're seeing a real pattern, or just a coincidence?

-  Is a new marketing campaign truly effective, or was last month's sales spike just luck?
-  Does a new drug actually work, or are the patient improvements due to random variation?

This deck introduces Hypothesis Testing: a powerful, structured framework used in science and business to make data-driven decisions under uncertainty.

The Language of Chance: A Quick Primer on Probability

Definition

Probability is a measure of the likelihood that a particular event will occur. It's a value between 0 and 1.

- **0:** The event is impossible.
- **1:** The event is certain to happen.

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

- **P(E)** is the probability of the event E
- **Favorable outcomes** are the outcomes we want
- **Total outcomes** are all possible outcomes

Classic Examples



- **Rolling a Dice:** The probability of rolling a 4 is **1/6 (or ~16.7%)**. There is one '4' out of six possible outcomes.



- **Drawing a Card:** The probability of drawing a red card from a standard 52-card deck is **26/52 (or 50%)**.

The Two Flavors of Statistics: Describing vs. Inferring

Statistics has two main branches that answer different questions.

Descriptive Statistics: What does our data show?

Summarizes the data you have.

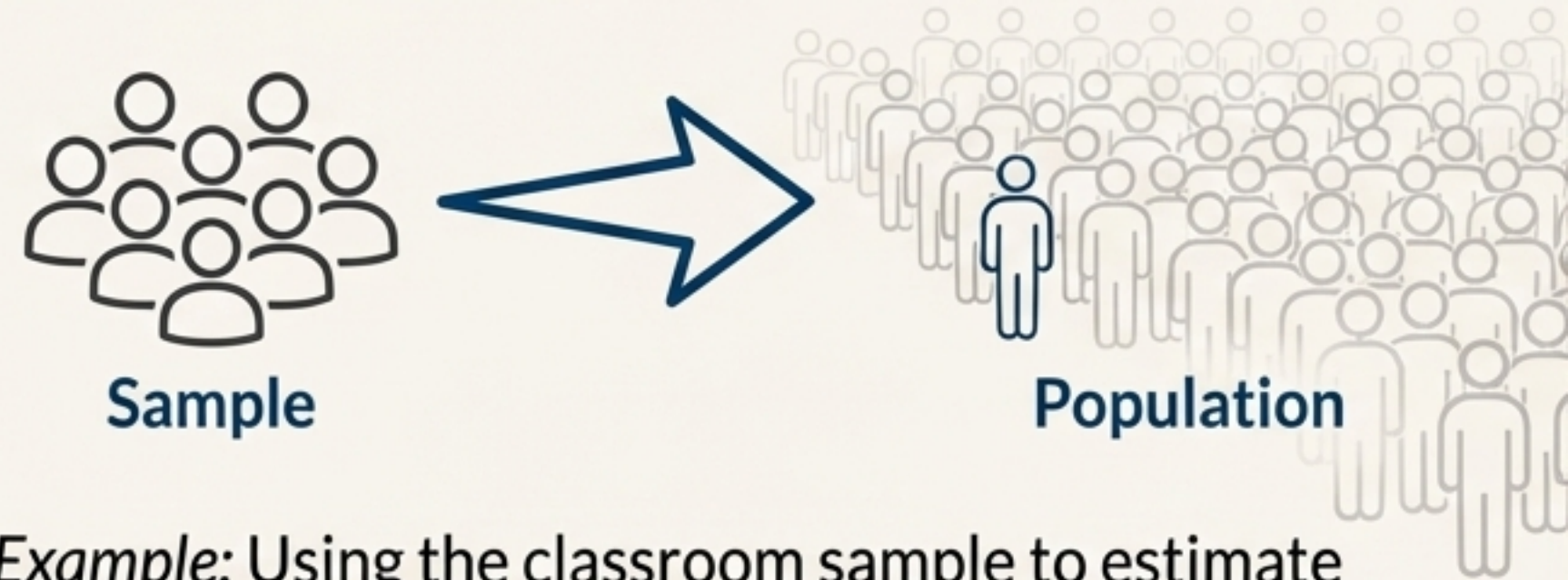


Example: Calculating the average height of students in *one classroom*.

Tools: Mean, Median, Mode, Standard Deviation.

Inferential Statistics: What can we guess about the data we *don't* have?

Uses a smaller **sample** to draw conclusions about a larger **population**.



Example: Using the classroom sample to estimate the average height of students in the *entire country*.

Hypothesis testing is a core tool of inferential statistics.
It gives us a formal process for making these educated guesses.

Our Framework: Hypothesis Testing as a Courtroom Trial

Think of hypothesis testing as a structured legal trial. It's a formal process to weigh evidence and make a decision.

The Core Idea

We start with a default belief (the 'status quo') and demand strong evidence to overturn it.



The Key Players in Our Trial



The Default Belief (The Accused):
The Null Hypothesis (H_0)



The New Claim (The Prosecution): The Alternative Hypothesis (H_1)



The Evidence:
Our Sample Data



The Standard of Proof:
The Significance Level (α)



The Verdict:
Our Statistical Decision

The Null Hypothesis (H_0): Innocent Until Proven Guilty

The **Null Hypothesis (H_0)** is the default assumption, the status quo, or the statement of “no effect.” It’s what we believe to be true before we even look at the evidence.

- In the Courtroom: “The defendant is **not guilty**.”
- In Business/Science:
 - “The new drug has **no effect**.”
 - “The average class score **is 70%**.”
 - “There is **no difference** between Group A and Group B.”



Our goal is not to prove H_0 , but to see if we have enough evidence to *reject* it.

The Alternative Hypothesis (H_1): The Prosecutor's Claim

The **Alternative Hypothesis (H_1 or H_a)** is the claim we want to test. It's the theory that challenges the status quo. It's what we hope to demonstrate with our data.

- In the Courtroom: "The defendant **is guilty**."
- In Business/Science:
 - "The new drug **works** (it has an effect)."
 - "The average class score **is greater than 70%**."
 - "There **is a difference** between Group A and Group B."



The burden of proof lies with the alternative hypothesis.

The Evidence: A Case Study on Student Scores

Scenario

The default belief (H_0) is that the average class score is exactly 70%. A teacher makes a claim (H_1) that the average is actually higher. We take a sample of student scores to investigate.

- H_0 : Average score = 70%
- H_1 : Average score > 70%

The Sample Data (The Evidence)

```
`scores = [72, 75, 80, 68, 71, 90, 85]`
```

The Statistical Test (The Analysis)

We use a one-sample t-test to compare our sample mean to the hypothesized population mean of 70.

```
from scipy import stats

scores = [72, 75, 80, 68, 71, 90, 85]
pop_mean = 70 # The value from our Null Hypothesis

# Perform the test
t_stat, p_value = stats.ttest_1samp(scores, pop_mean)

print(f"p-value: {p_value:.4f}")
```


Weighing the Evidence: What is a P-value?



The **P-value** is the probability of seeing our sample results (or something even more extreme) **if the Null Hypothesis (H_0) were actually true.**

Translation (Courtroom Analogy): “How likely is it that we’d find this evidence if the defendant were truly innocent?”

- **A low P-value** means our evidence is very surprising and unlikely if H_0 is true. It casts doubt on the “innocence” of H_0 . ?
- **A high P-value** means our evidence is quite plausible and not surprising under H_0 . It doesn’t give us a reason to doubt H_0 . ✓

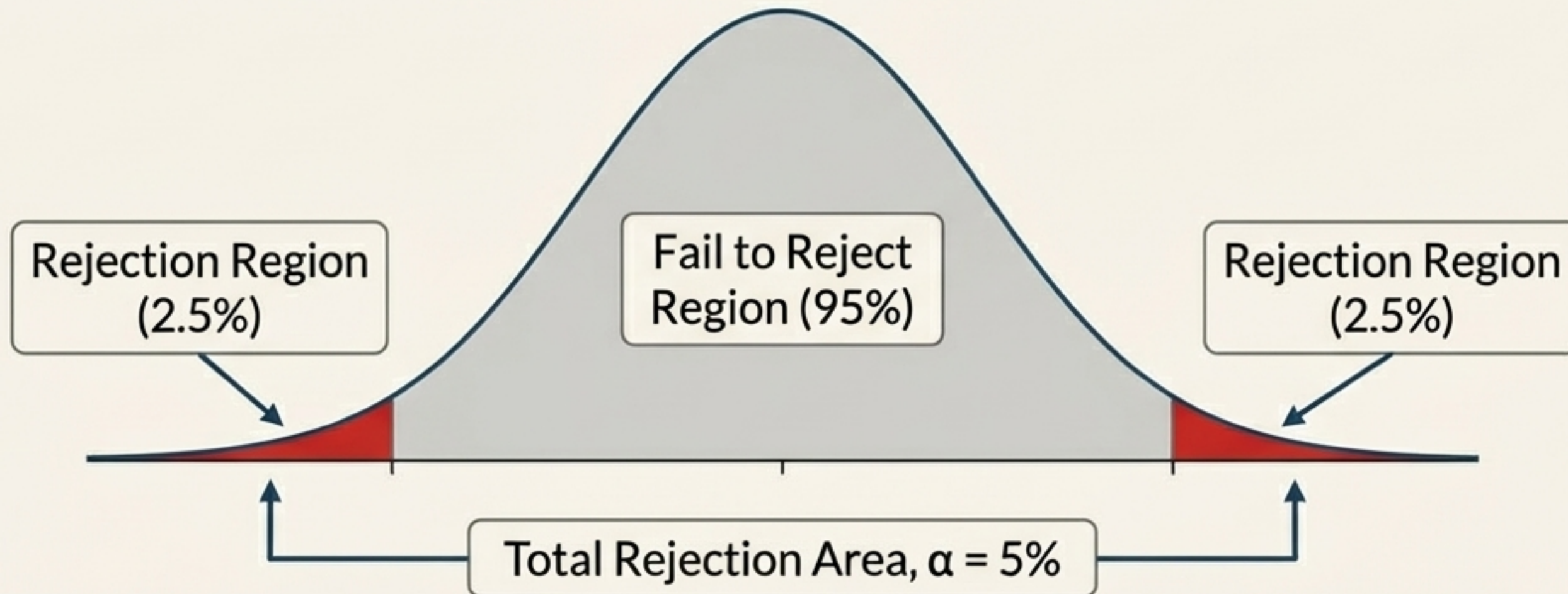
Important: The p-value is NOT the probability that the null hypothesis is true.

The Standard of Proof: Significance Level (α)

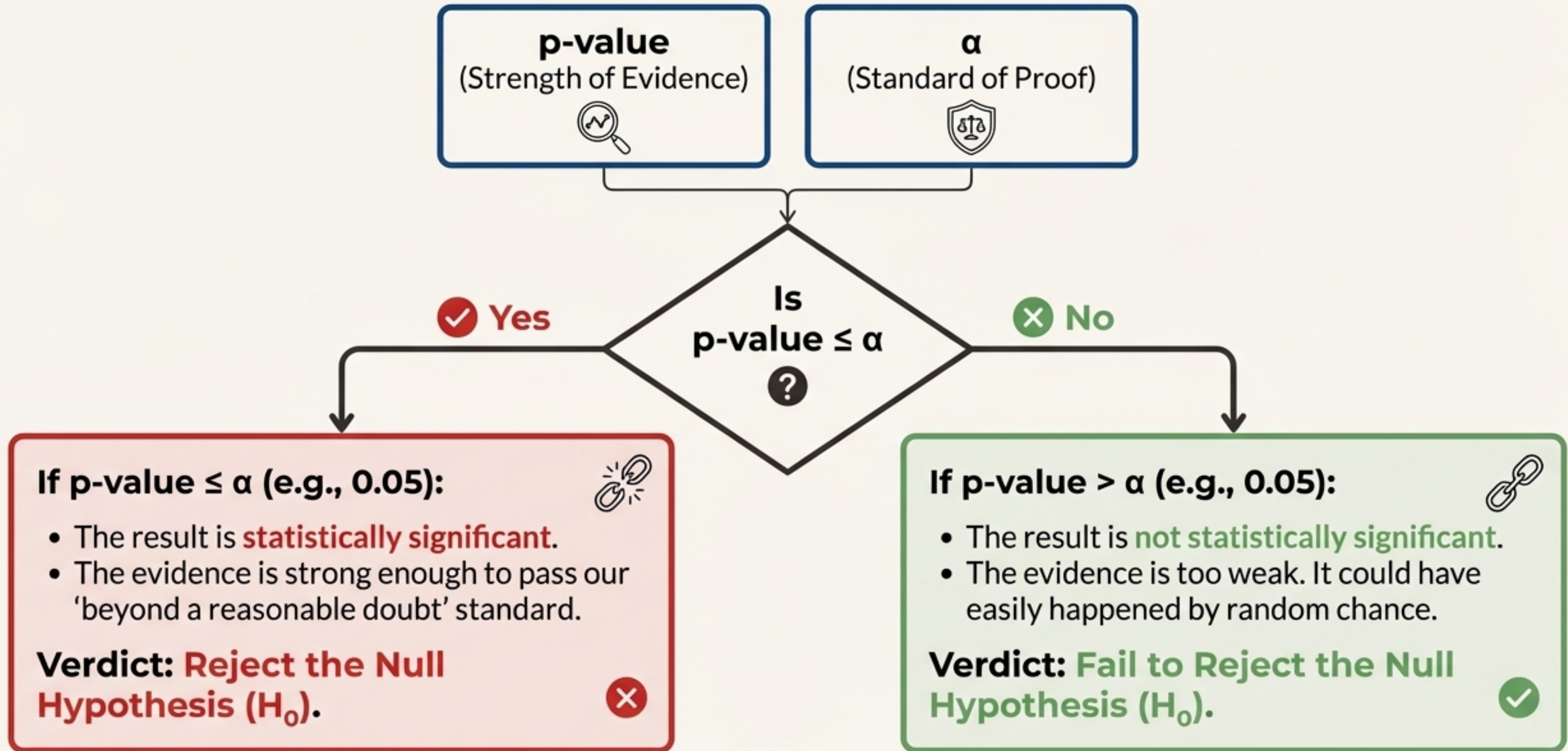
Before we even run our test, we set a **Significance Level, denoted by α (alpha)**.

This is our threshold for 'beyond a reasonable doubt.' It's the probability of rejecting the null hypothesis when it is actually true (we'll call this a Type I Error later).

The universal standard in science and industry is $\alpha = 0.05$ (or 5%). This means we are willing to accept a 5% risk of incorrectly rejecting the null hypothesis.



The Verdict: The Simple Rule for Making a Decision



Case Closed: The Verdict for the Student Scores

Recall our Scenario:

- H_0 : Average score = 70%
- H_1 : Average score > 70%
- Our chosen $\alpha = 0.05$

The Test Result:

Running the Python code from before gives us: **p-value = 0.0349**

The Decision:

0.0349 $\leq$ 0.05? **Yes. ✓**

The Conclusion:

Our p-value is less than our significance level. The result is statistically significant.

We reject the Null Hypothesis. 

There is sufficient statistical evidence to suggest that the true average class score is higher than 70%.

****Crucial Wording**:** We never 'accept the alternative hypothesis.' We simply state that we have enough evidence to reject the null.

But What if the Verdict is Wrong? The Two Types of Error

Our statistical verdict is based on probability, not certainty. We can make mistakes. In hypothesis testing, there are two specific ways we can be wrong. Understanding these errors is critical for interpreting results and managing risk in the real world.

	Reality: H_0 is True (Innocent)	Reality: H_0 is False (Guilty)
We Reject H_0 (Verdict: Guilty)	Type I Error (False Positive)	Correct Decision (True Positive)
We Fail to Reject H_0 (Verdict: Not Guilty)	Correct Decision (True Negative)	Type II Error (False Negative)

Type I Error (False Positive): Convicting the Innocent

Definition: A Type I error occurs when we **reject a null hypothesis that was actually true**.

In Simple Terms: We find a “significant” result that was just due to random chance. We raise a false alarm.

The Analogies



Courtroom: An innocent person is convicted of a crime.



Medicine: A medical test tells a healthy person they have a disease.



Business: We conclude a new marketing campaign works when it had no real effect.

The probability of making a Type I error is equal to our significance level, α .

Type II Error (False Negative): Letting the Guilty Go Free

Definition: A Type II error occurs when we **fail to reject a null hypothesis that was actually false**.

In Simple Terms: We miss a real effect or a true pattern that was there. It's a missed opportunity.

The Analogies



Courtroom: A guilty person is set free.



Medicine: A medical test fails to detect a disease in a sick person.



Business: We abandon a promising new drug because our initial test failed to show its (real) effectiveness.

There is always a trade-off: decreasing the **chance of a Type I error** often increases the chance of a Type II error, and vice-versa.

The Data Scientist's Toolkit:

A Principled Approach to Uncertainty

Hypothesis testing isn't about finding absolute truth. It's a disciplined framework for using limited data to navigate an uncertain world. It forces us to define our assumptions, set a standard of proof, and make the best possible decision, all while understanding the risks involved.

Term	Role in the "Trial"	Purpose
 Hypothesis Testing	The Entire Court Case	A formal process to test a claim.
 Null Hypothesis (H_0)	The Defendant ("Innocent")	The default assumption of "no effect."
 Alternative Hyp. (H_1)	The Prosecutor's Claim ("Guilty")	The new theory you are testing.
 P-value	Strength of the Evidence	Measures how surprising your data is if H_0 is true.
 Significance (α)	Standard of Proof	The cutoff for your decision (usually 0.05).
 Type I Error	Convicting the Innocent	A "false alarm."
 Type II Error	Freeing the Guilty	A "missed opportunity."